Jairus Tanaka

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Education

Geneva College Beaver Falls, PA
BS Computer Science, Minor in Software Development Aug. 2024-May. 2027
All Saints Academy Silverdale, WA
High School Diploma Sept. 2020-June 2024

Technical Skills

Programming: Go, C, C++, C#, TypeScript, Java, Go, SQL, Rust, WebAssembly, AssemblyScript, Python, x86 Assembly, RISC-V, ARM64/32, Xtensa
Systems & Runtime: Compiler Design, JIT and AOT Compilation, SIMD/SWAR, Sandboxing, Virtual Machines, Emscripten, Runtime Optimization, Memory Management, WASI
Backend & APIs: Nodejs, Bun, Express, Fastify, GraphQL, REST
Databases & Storage: MySQL, Redis, MongoDB, SingleStore, Scylla, Dgraph
Tools & Infra: Git, Linux, CI/CD, GitHub Actions, Docker, WireGuard, Postman, CLI Development, Benchmarking, Performance Profiling

Projects

Wago June 2026 – Present
Architected and developed a pure-Go WebAssembly runtime from scratch featuring a custom single-pass compiler, advanced compiler optimizations, efficient memory management, and extensible runtime APIs. Implemented native code generation backends targeting x86-64, ARM64, and RISC-V. Across 145 real-world workloads, achieved faster execution than LLVM-compiled code while delivering up to 68× faster end-to-end compilation and execution, as measured by wasm.fyi

High-Performance JSON May 2021 – Present
Designed and implemented a high-performance JSON serialization and parsing library for AssemblyScript, achieving multi-GB/s throughput through custom algorithms, SIMD/SWAR optimizations, compiler integrations, specialized code generation, and custom memory allocation. Adopted in production by multiple companies (Impart Security, Steer Protocol, Massa Labs, Chipotle, BMW, Sony, Bifrost), surpassed 300k NPM downloads, and deployed to over 10 million BMW vehicles.

The AssemblyScript Project October 2026 – Present
Maintainer of AssemblyScript, a TypeScript-like programming language that compiles to highly optimized WebAssembly. Contributed and lead compiler development, runtime optimizations, standard library improvements, and ecosystem tooling.

Relevant Experience

Impart Security Hybrid, Full-time
Software, Systems, and Performance Engineer May 2026 – Present
Spearheaded optimization work on our inspectors which monitor ingress data for malicious content or bad data. Redesigned each with custom algorithms that utilize SIMD, SWAR, parallel computation, threads, etc. Specced and designed new a new architecture which offered better parallelism and bumped throughput by over 3x. Found and implemented fixes for multiple security vulnerabilities, RFCs, and more. Created custom garbage collectors, allocators, runtimes, languages, and libraries, essentially owning the entire stack. Since joining, our raw inspection throughput has been bumped by 428% on single-core, over 1,487% on multi-core, and our running-costs more than cut in half.

Hypermode Inc Remote, Part-time
Software and Performance Engineer June 2024 – July 2025
Authored core libraries and frameworks including an extensive testing framework, cli, compiler transforms, and data parsing and analysis libraries. Re-engineered the architecture that we use for passing data which allowed us to do so with much lower latency and overhead than before. Maintained Modus, our underlying custom WebAssembly runtime and took over development of its corresponding Go and AssemblyScript SDKs. Revised and updated the documentation to improve onboarding and developer experience.

Steer Protocol Remote, Part-time
Software Engineer July 2023 – Aug. 2024
Designed a WebAssembly engine for secure execution of on-chain user-created programs. Built and maintained extremely low latency data analysis pipelines, built data connector libraries, and designed trading strategies running in production. Developed supporting libraries including a RLP implementation (the format Ethereum uses), an on-chain http client, an IEEE-754-compatible integer-based float implementation akin to softfloat, and more.

Leadership

World Wide Web Consortium The Web, Remote
WebAssembly CG Dec. 2021 – Present
Participated in the W3C WebAssembly Community Group, contributing to discussions centering around WebAssembly proposals, standards, tooling, and ecosystem development. Occasionally offered practical feedback based upon my experience both using and creating WebAssembly runtimes, compilers, and production systems.

Geneva College Beaver Falls, PA
Student Instructor Aug. 2025 – Present
Led classes of 22 and 14 students respectively. Supported them by teaching study skills, good habits, college readiness, and foundational coursework. Led class sessions, discussions, hosted review and study sessions, and provided extensive one-on-one tutoring to many students.